Random Forest Analysis of Diabetes Health Indicators

ST563: Random Forest vs. Logistic Regression and KNN

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Random forest (RF) models are fitted to the same three datasets and the same 80/20 train–test split (seed 42) as the earlier logistic-regression (LR) and KNN analysis, so results are directly comparable.

- **Balanced** binary: 1 = diabetes, 0 = no diabetes / prediabetes.
- **Imbalanced** binary: 1 = prediabetes or diabetes, 0 = no diabetes.
- **3-class**: 0 = no diabetes, 1 = prediabetes, 2 = diabetes.

1 Model parameters

Same hyperparameters for all three forests (defaults except `ntree` and `importance`).

Hyperparameter	Setting
Number of trees (ntree)	500
mtry (candidates per split)	4 (default = floor(sqrt(21)))
nodesize (min terminal-node size)	1 (default)
maxnodes (tree-depth limit)	none (fully grown trees)
sampsize (bootstrap sample size)	n (one bootstrap draw per tree, with replacement)
replace (bootstrap resampling)	TRUE (default)
Split criterion	Gini impurity
importance	TRUE (permutation + Gini)
Type	classification

2 Prediction results

2.1 Binary (balanced and imbalanced)

Dataset	Accuracy	Precision	Recall	Specificity	F1	AUC
Balanced	0.7415	0.7240	0.7806	0.7024	0.7512	0.8087
Imbalanced	0.8425	0.5693	0.1549	0.9771	0.2436	0.7873

2.2 3-class

Class	Precision	Recall	F1	OVR.AUC
No diabetes (0)	0.8531	0.9813	0.9127	0.7858
Pre-diabetes (1)	0.0000	0.0000	0.0000	0.6172
Diabetes (2)	0.5489	0.1487	0.2340	0.7982

Accuracy	Macro.F1	Mean.OVR.AUC
0.8415	0.3822	0.7337

3 Comparison with LR and KNN

3.1 Binary

model	dataset	accuracy	precision	recall	F1	AUC
Logistic regression	Balanced	0.7418	0.7318	0.7633	0.7472	0.8152
Logistic regression	Imbalanced	0.8433	0.5651	0.1860	0.2799	0.8055
KNN	Balanced	0.7361	0.7156	0.7837	0.7481	0.8089
KNN	Imbalanced	0.8430	0.6056	0.1171	0.1962	0.7989
Random forest	Balanced	0.7415	0.7240	0.7806	0.7512	0.8087
Random forest	Imbalanced	0.8425	0.5693	0.1549	0.2436	0.7873

3.2 3-class

model	accuracy	macro_F1	mean_OVR_AUC
Multinomial LR	0.8412	0.3917	0.7660
KNN	0.8406	0.3498	0.7520
Random forest	0.8415	0.3822	0.7337

Random forest matches the ranking ability of LR and KNN on the binary tasks (AUC around 0.79–0.81) and, unlike them, predicts a small non-zero share of the rare prediabetes and diabetes classes in the 3-class task.